DNCS24

16.3.2026

Microsoft Azure User and Virtual Machine Management



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1 Introduction

This report describes the basic use of Microsoft Azure for managing users and virtual machines. The aim of the exercise is to practice creating new users in Microsoft Entra ID, deploying a new virtual machine, and managing access permissions for that machine. Report also demonstrates how Azure AD authentication can be used to allow users to connect to a virtual machine with SSH.

2 Azure instructions

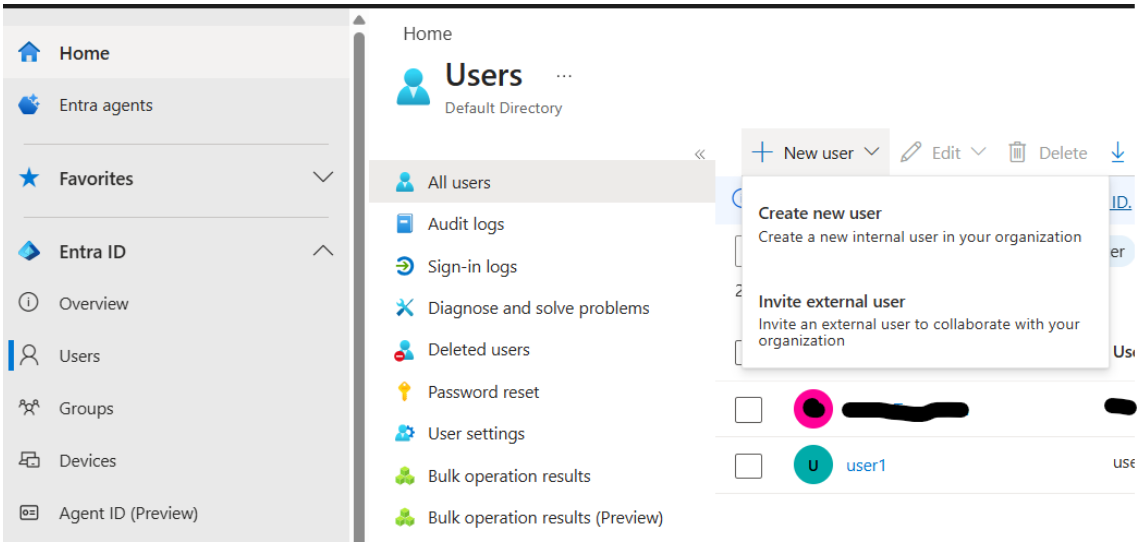
This section includes instructions for:

- Creating new users inside the organization
- Deploying new virtual machines
- How to control user permissions who can access virtual machine
- How users can connect to virtual machine

2.1 Creating new user

You can create new users for your organization in two different places. You can navigate to:

- Azure Portal: **portal.azure.com**
  - choose **Microsoft Entra ID** from Azure services or type it in to search bar. From **Default Directory** page press **+ Add** and **User -> Create new user**
- Microsoft Entra: **entra.microsoft.com**
  - Press **Users** from the left and then **+ New user -> Create new user**

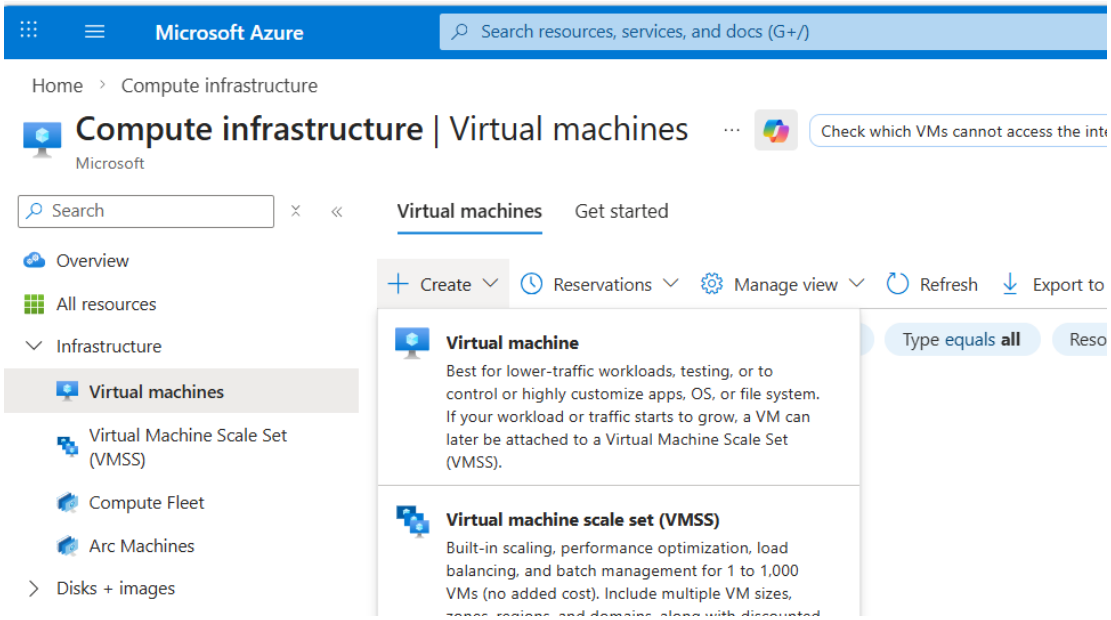


Then add the user with the information you want to give to the user.

2.2 Creating new Virtual Machine

Navigate to **Virtual Machines** in Azure Portal from the left-hand side portal menu or typing to search bar **virtual machines.**

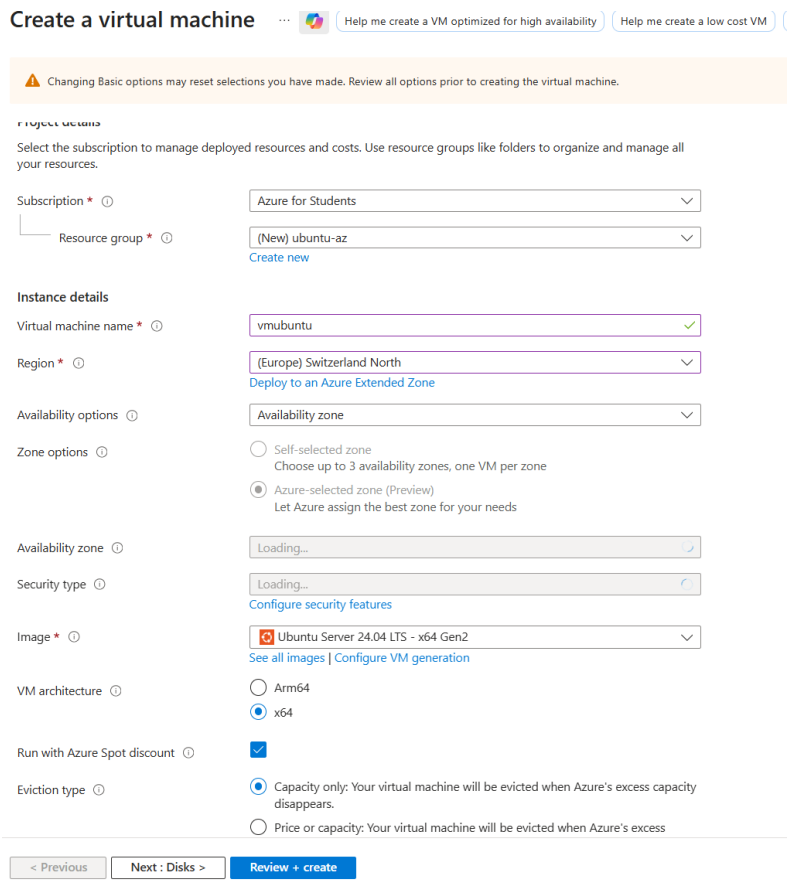
- press **+ Create -> Virtual machine**



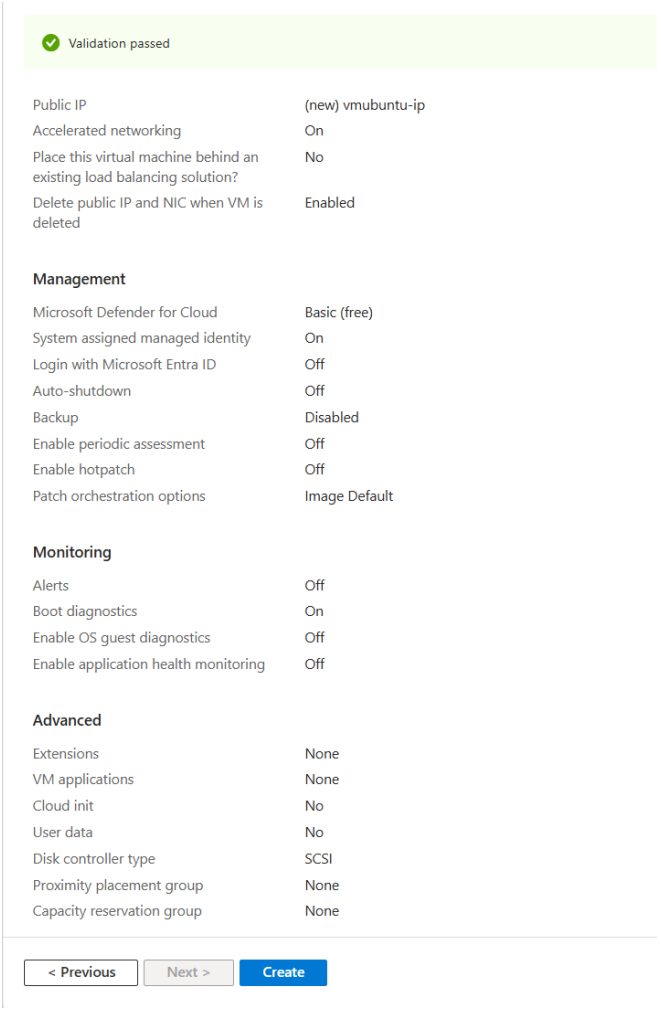
When creating new virtual machine you need to set up:

- Subscription
- resource group
- Virtual machine name
- Region
- Image
- Size
- Admin account -> Password
    - Username
    - Password
- public inbound ports (SSH)

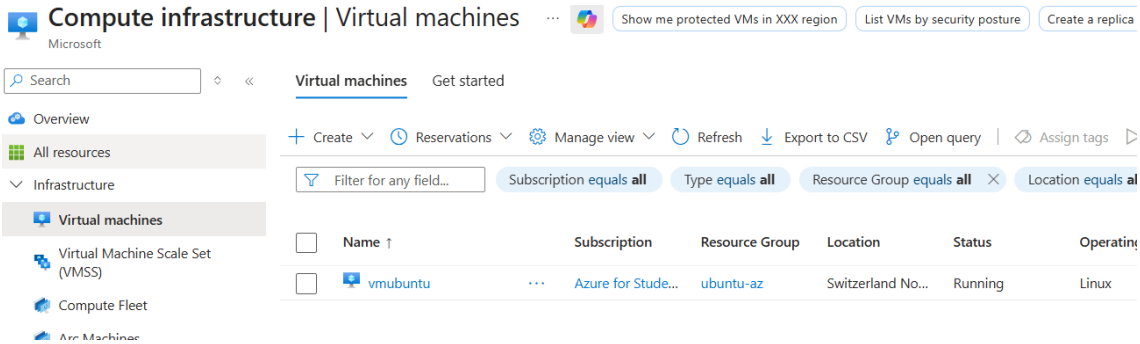
- Disks -> you can change disk type
- Networking -> virtual network
    - Create new or select from the list
- Management -> enable tab “login with Microsoft Entra ID”



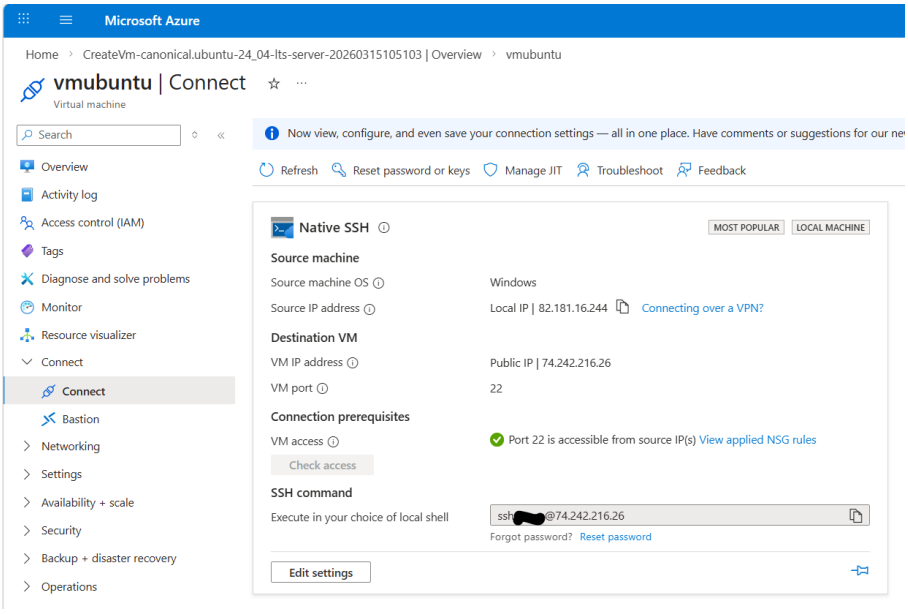
When you have correct setup, you should get green banner at the last setup page which says, “Validation passed” and then click **Create**.



Now you should find your new virtual machine from the Azure **Virtual machines.**

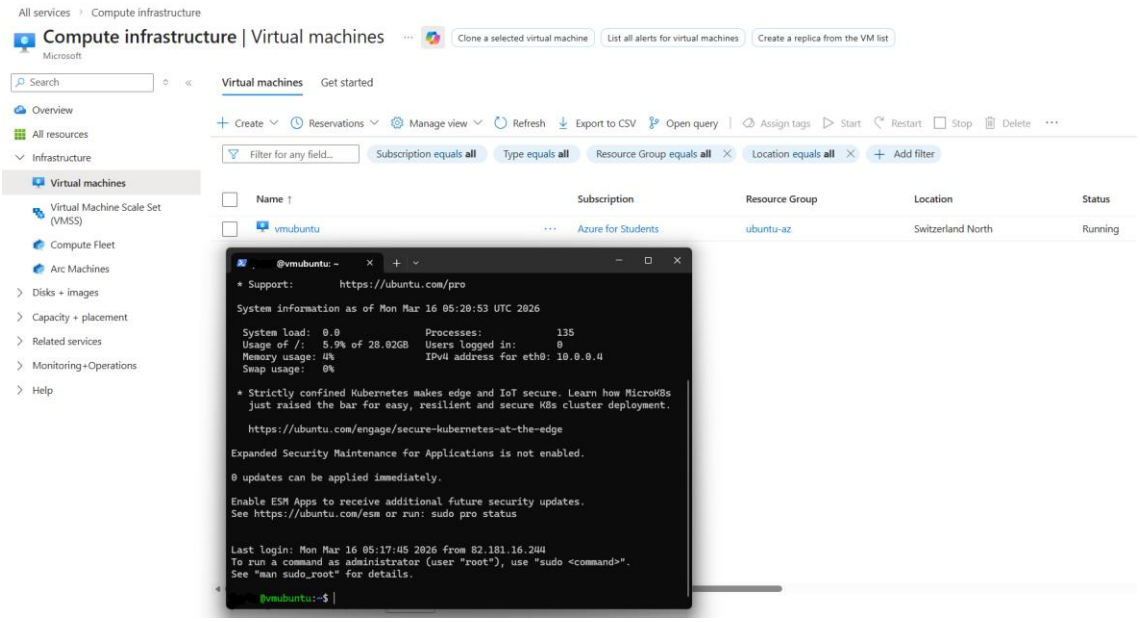


To test your new virtual machine, select it from the list and press **Start** from the top tool bar. Once it is running, select **Connect** and you should see IP address of the machine and SSH command you can use.



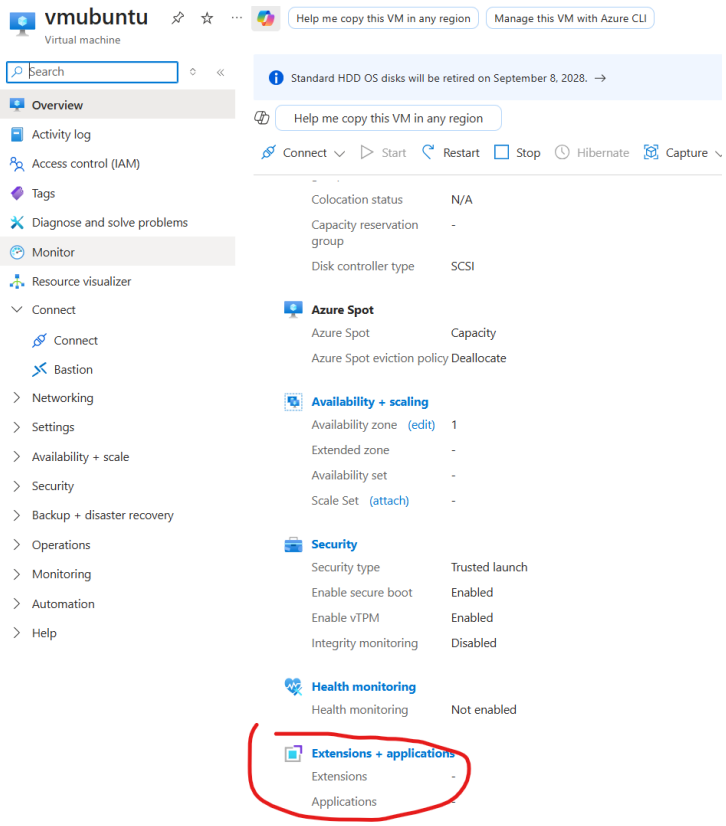
On your own computer or inside the Azure Portal, open PowerShell and connect with SSH command:

ssh username@ipaddress

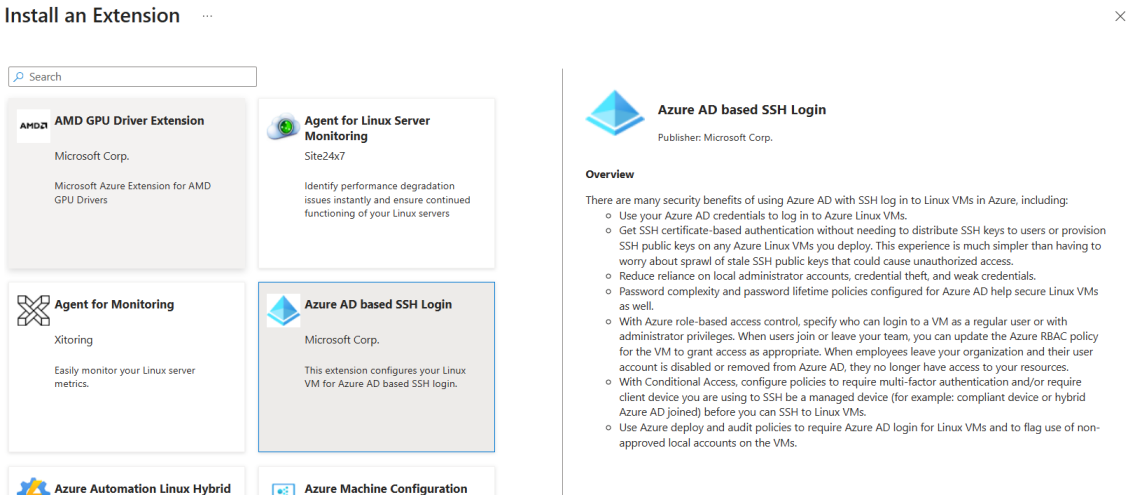


2.3 Accessing virtual machine with Azure AD (Entra) user

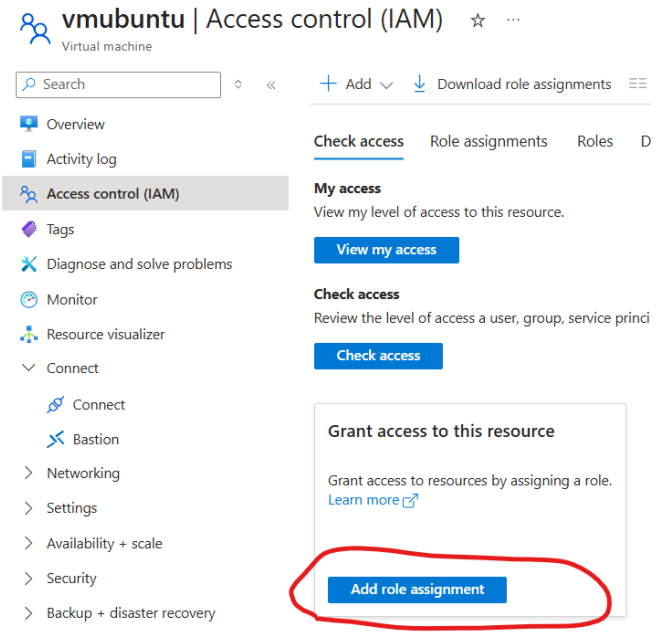
First you need to install an extension for the virtual machine, which enables Azure AD based Login. Go to your virtual machine in Azure and scroll down or search from the left: **Extensions + applications**.



Install extension called **Azure AD based SSH Login**.

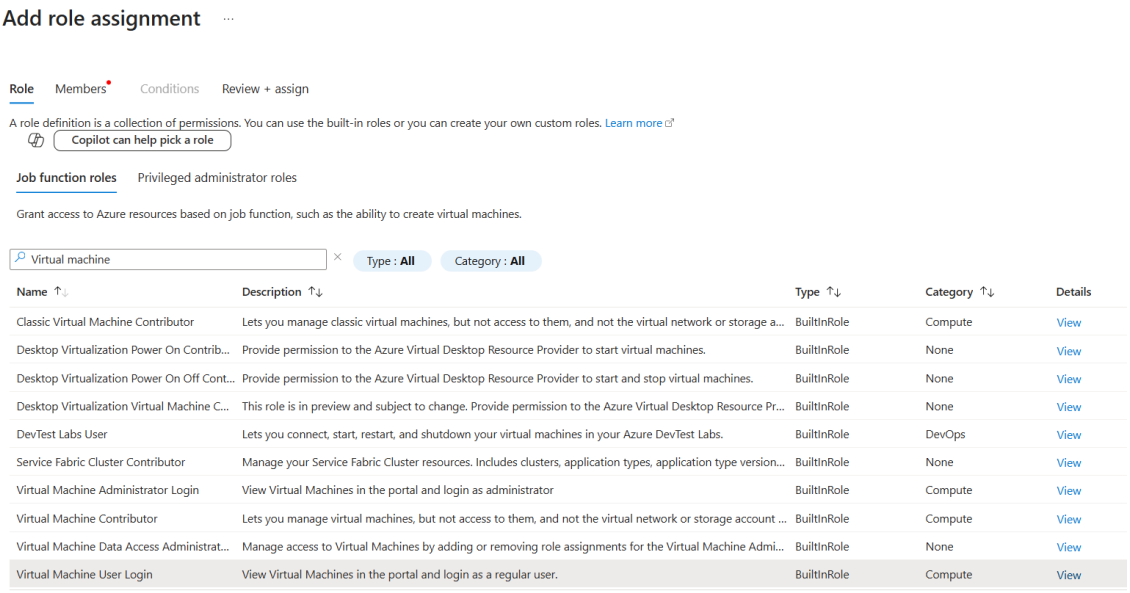


After the extension is installed, go to your virtual machine and from the left select **Access control (IAM).**

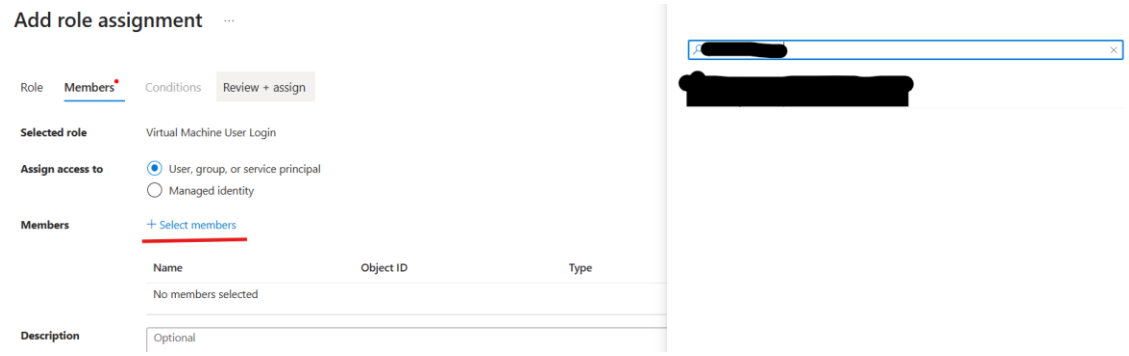


Press **Add role assignment.**

Type in the search bar “virtual machine” and then from the list, select **Virtual Machine User Login.**

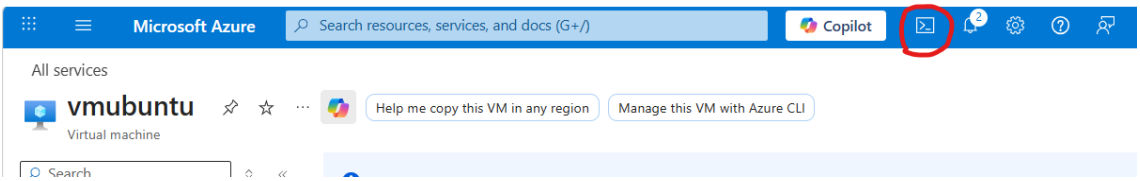


Then you can select members who can access this virtual machine.

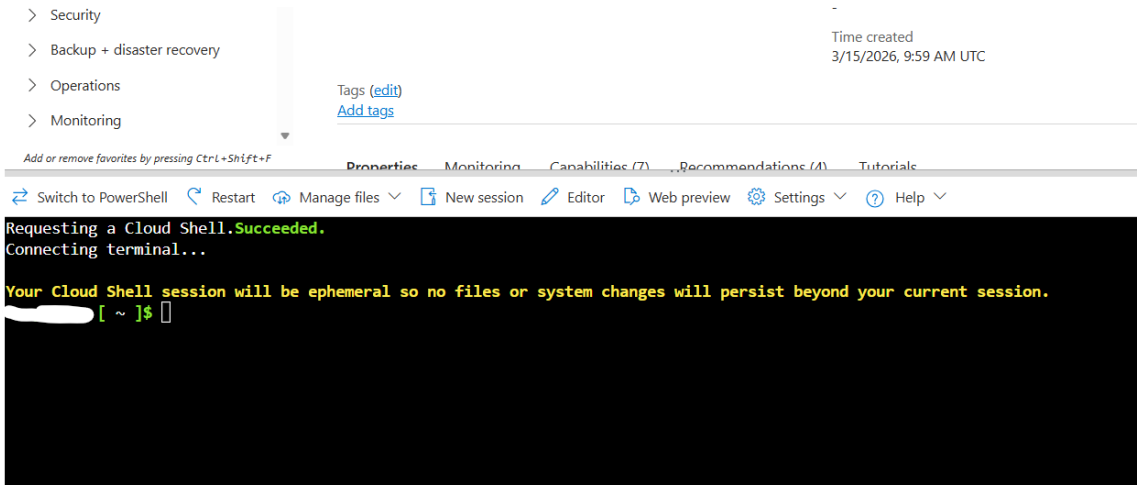


Press **+ Select members** and look for the user you want to add. Now the user should be able to connect to machine.

Login to Azure Portal with the connected user and test the connection to virtual machine with **Cloud Shell.** Cloud shell can be accessed with the button on top bar of the Azure Portal.



Terminal will open to the bottom of the screen.



Here you can connect to the virtual machine with command:

az ssh vm --name VMNAME --resource-group RESOURCEGROUP

Where:

- VMNANE = your virtual machine name
- RESOURCCEGROUP = Resource group in which the machine belongs to. You can check this from the virtual machines **Overview** tab.

